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COMPUTER SECURITYWK1

1.Scenario Analysis

A company's IT department performs regular file integrity checks to protect important files from unauthorized changes. The system creates a unique digital hash for each critical file and stores the original hashes in a secure location. During scheduled checks, the system calculates new hashes and compares them with the stored values. If the hashes match, the files are considered unchanged. However, if a hash is different, it indicates that the file may have been modified, corrupted, or accessed by an unauthorized person. For example, if an employee's financial report is changed without permission, the integrity-checking system will detect the difference and generate an alert. The IT security team can then investigate the incident, identify the cause, restore the original file from a backup, and review access logs. This security measure helps prevent data tampering, detect attacks early, and maintain the accuracy and reliability of important information.

2. Concept Research:

Confidentiality is a key principle of information security that ensures sensitive information is only accessed by authorized individuals. Its main role is to prevent unauthorized users from viewing, copying, or sharing private data. Organizations protect confidentiality through methods such as passwords, encryption, access controls, and user authentication. For example, a hospital must keep patients' medical records confidential and only allow authorized staff to access them. If confidentiality is compromised, sensitive information may be stolen, misused, or exposed. Maintaining confidentiality helps organizations protect personal and business information, build trust with users, comply with data protection laws, and reduce the risk of security breaches.

3. Tool Practice

I used Wireshark to simulate a basic network security monitoring process. Wireshark allowed me to capture and examine network packets being transmitted between devices. I observed information such as source and destination IP addresses, protocols, and packet details. This activity helped me understand how network traffic can be monitored to identify unusual or suspicious communication. I learned that protocols such as TCP, UDP, and DNS can provide useful information when investigating network activity. The exercise also showed me that Wireshark can help security professionals detect potential threats, troubleshoot network problems, and analyze suspicious traffic. Overall, the practice improved my understanding of network security and the importance of monitoring network activity.

4. Security Diagram

The CIA Triad is a fundamental security concept made up of Confidentiality, Integrity, and Availability. In my Canva diagram, these three principles are connected to show how they work together to protect information. Confidentiality ensures that sensitive information can only be accessed by authorized users, using methods such as passwords and encryption. Integrity ensures that information remains accurate and is not changed without permission, using tools such as file integrity checks and backups. Availability ensures that authorized users can access systems and information when needed through reliable networks, backups, and maintenance. The CIA Triad helps organizations protect their data from unauthorized access, accidental changes, and service interruptions.